Han Joon Byun

Researcher in Machine Learning, Optimization, and Finance

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RESEARCH PROFILE

Researcher working at the intersection of machine learning, mathematical optimization, and finance, with a focus on tabular data and time-series modeling. Current research combines evolutionary computation and Transformer architectures to improve predictive performance, computational efficiency, and interpretability. Brings rigorous mathematical training and applied experience spanning financial factors, credit-risk prediction, and interpretable AI.

Research areas: Tabular machine learning · Time series · Evolutionary computation · Transformer models · Financial AI · Interpretable machine learning

EDUCATION

Ph.D. in Computer Science and Engineering — Expected 2027
Seoul National University · Optimization Lab

M.S. in Computer Science and Engineering — 2024
Seoul National University · Optimization Lab

B.S. in Mathematics — 2018
New York University
Advanced study included Probability Theory, Statistics, and Stochastic Calculus.

RESEARCH & APPLIED EXPERIENCE

Agency for Defense Development · Joint Research Project — 2026
- Contributed to research on interpretable modeling, emphasizing methods that make model behavior easier to understand.

PFCT · Joint Research Project — 2025
- Developed a default-prediction model using tabular Transformer architectures for structured financial data.

Think Pool · Joint Research Project — 2022–2024
- Developed AI-driven financial factors, applying machine-learning methods to quantitative finance.

Republic of Korea Air Force · Interpretation Officer — 2019–2022
- Served as an interpretation officer and established a long-term career direction at the intersection of computer science and finance.

TEACHING EXPERIENCE

Head Teaching Assistant · Data Structures — Spring 2023 · Fall 2024 · Fall 2025
Seoul National University

Head Teaching Assistant · Algorithms — Fall 2023 · Spring 2025 · Spring 2026
Seoul National University

Prepared course content, coordinated teaching support, answered student questions, and graded assignments and examinations. Maintained an approachable, hands-on presence to help students build confidence in core problem-solving concepts.

RESEARCH TRAJECTORY

Neural Network Foundations

Studied CNN and ANN pruning and transfer learning to understand model learning and efficiency.

Combinatorial Optimization

Helped develop a bespoke Transformer that frames the Traveling Salesperson Problem through language modeling.

Financial & Tabular AI

Applies optimization and neural methods to improve performance, efficiency, and interpretability.

PUBLICATIONS

1. **RTSC: Return-Aware Token-Space Contrastive Transformer for Cross-Sectional Stock Return Prediction**
   Hanbin Koo, **Han Joon Byun**, and Byung-ro Moon
   Submitted to ICAIF 2026 · **Under review**

2. **Certificates, Declarations, and Abstention After Offline Policy Selection: Two Synthetic Operating-Characteristic Calibrations**
   Kihun Rhee and **Han Joon Byun**
   Submitted to Transactions on Machine Learning Research (TMLR) · **Under review**

3. **Label-Noise Decay Confounds Loss-Based Drift Attribution: A False-Positive Channel and a Noise-Aware Repair**
   Huh Joon and **Han Joon Byun**
   Submitted to Transactions on Machine Learning Research (TMLR) · **Under review**

4. **Multi-Objective Evolution of Patch Tokens for Tabular Transformers**
   Han Joon Byun, Joon Huh, Jieun Yook, and Byung-ro Moon
   Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '26), 2026
   doi.org/10.1145/3795095.3805076

5. **Evolutionary Feature Construction for Tabular Transformers via Gradient-Enhanced Attention Rollout**
   Han Joon Byun, Jieun Yook, Hanbin Koo, and Byung-ro Moon
   Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '26 Companion), 2026
   doi.org/10.1145/3795101.3805282

6. **CycleFormer: TSP Solver Based on Language Modeling**
   Jieun Yook, Junpyo Seo, Joon Huh, **Han Joon Byun**, and Byung-ro Moon
   arXiv preprint, 2024
   arxiv.org/abs/2405.20042

7. **Evolutionary Pruning of Deep Convolutional Networks by a Memetic GA with Sped-Up Local Optimization and GLCM Energy Z-Score**
   Ha Hyun Cho, **Han Joon Byun**, Min Kee Kim, Joon Huh, and Byung-Ro Moon
   Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '23 Companion), 2023
   doi.org/10.1145/3583133.3590604